

Install libraries

```
In [ ]: !pip install -q transformers accelerate bitsandbytes sentence-transformers faiss-cpu pypdf
```

```
60.7/60.7 MB 16.3 M
B/s eta 0:00:00
23.8/23.8 MB 74.5 M
B/s eta 0:00:00
335.6/335.6 kB 23.9
MB/s eta 0:00:00
```

Chunk doc

```
In [ ]: from pypdf import PdfReader

def load_pdf(path):
    reader = PdfReader(path)
    text = ""
    for page in reader.pages:
        text += page.extract_text() + "\n"
    return text

def chunk_text(text, chunk_size=500, overlap=100):
    chunks = []
    for i in range(0, len(text), chunk_size - overlap):
        chunks.append(text[i:i+chunk_size])
    return chunks

text = load_pdf("LIC_policy.pdf") # upload in Colab
chunks = chunk_text(text)

print("Total chunks:", len(chunks))
```

Total chunks: 152

Embeddings + Vector Store

```
In [ ]: from sentence_transformers import SentenceTransformer
import faiss
import numpy as np

embed_model = SentenceTransformer("all-MiniLM-L6-v2")

embeddings = embed_model.encode(chunks, show_progress_bar=True)

dimension = embeddings.shape[1]
index = faiss.IndexFlatL2(dimension)
index.add(np.array(embeddings))
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status	
embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Retriever

```
In [ ]: def retrieve(query, k=3):  
        q_emb = embed_model.encode([query])  
        distances, indices = index.search(np.array(q_emb), k)  
        return [chunks[i] for i in indices[0]]
```

Load SLM

```
In [ ]: from transformers import AutoTokenizer, AutoModelForCausalLM  
import torch  
  
model_name = "Qwen/Qwen2.5-1.5B-Instruct" # lightweight for Colab  
  
tokenizer = AutoTokenizer.from_pretrained(model_name)  
model = AutoModelForCausalLM.from_pretrained(  
    model_name,  
    device_map="auto",  
    torch_dtype=torch.float16  
)
```

`torch\_dtype` is deprecated! Use `dtype` instead!

Generator

```
In [ ]: def generate_answer(query):
        docs = retrieve(query, k=3)
        context = "\n\n".join(docs)

        prompt = f"""
        You are a helpful assistant. Answer ONLY from the context.

        Context:
        {context}

        Question:
        {query}

        Answer:
        """

        inputs = tokenizer(prompt, return_tensors="pt").to(model.device)

        outputs = model.generate(
            **inputs,
            max_new_tokens=200,
            temperature=0.3
        )

        return tokenizer.decode(outputs[0], skip_special_tokens=True)
```

Test the model

```
In [ ]: query = "What is free look period?"  
        print(generate_answer(query))
```

You are a helpful assistant. Answer ONLY from the context.

Context:

endorsements or

modifications agreed to or issued by the Corporation.

18. Free Look Period is the period of 15 days (30 days if this policy is purchased online) from the date of receipt of the Policy Document by the Policyholder to review the terms and conditions of this policy and where the Policyholder disagrees to any of those terms and conditions, he/ she has the option to return this policy.

19. IRDAI means Insurance Regulatory and Development Authority of India earlier called as Insura

der and as specified in the

Schedule, commencing from the Date of commencement of policy to the date of Vesting.

15. Discharge form is the form to be filled by Annuitant /Claimant to claim the Surrender/Death benefit under the policy.

16. Due Date means a fixed date on which the annuity is due and payable.

17. Endorsement means conditions attached/ affixed to this Policy incorporating any amendments or modifications agreed to or issued by the Corporation.

18. Free Look Period is the pe

tion of interest rate for policy loan shall be subject to prior approval of IRDAI.

In case of QROPS, above policy loan provisions shall be subject to Rules and Regulations of HMRC in this regard.

6. Free Look period:

During the Free Look period of 15 days (30 days if this policy is purchased online) from the date of receipt of the Policy Document by the Policyholder, if the Policyholder is not satisfied with the Terms and Conditions of the policy, he/she may return the policy to the

Question:

What is free look period?

Answer:

The free look period refers to the 15-day window (or 30 days if purchased online) during which the policyholder can review the terms and conditions of the insurance policy before making a decision to either accept or return it. This allows the policyholder to ensure that they understand all aspects of the policy without being bound until the end of the free look period. If the policyholder decides to return the policy within this timeframe, they must do so before the end of the free look period. The purpose of the free look period is to provide an opportuni

ty for the policyholder to make an informed decision about whether to continue with the policy or seek alternative options.

In []: